



SPYWOLF

Security Audit Report

Audit prepared for
PerShare

Completed on
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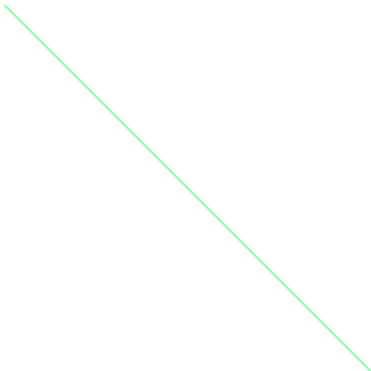
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PROJECT INFORMATION

- **Project Name:** PerShare (PerShare V1)
Token / Short Name: SHARE
Category: On-chain collective payment pool — a group-buy escrow that lets 2 to 50 members pool stablecoins toward a shared target, validate collectively, and redistribute presale tokens pro-rata.
- **Network:** BNB Smart Chain (BSC Mainnet, chainId 56). Testing and verification were configured for BSC Testnet (chainId 97) ahead of mainnet deployment.
Token Standard: BEP-20 / ERC-20.
Language: Solidity ^0.8.20.
Compiler: solc 0.8.20, viaIR enabled, optimizer enabled at 200 runs.
Framework: Hardhat.
Libraries: OpenZeppelin Contracts v5.6.1 — ReentrancyGuard, Pausable, and Ownable.
- **Files in Scope:** contracts/PerShareV1.sol (the PerShare contract), totalling 541 lines of code.
Out of Scope: MockUSDT.sol, MockToken.sol, and MockPresaleToken.sol, which are test-only mocks explicitly flagged as not for mainnet deployment.
- **Core Functionality:** Phase 1 handles stablecoin collection, collective member validation, and transfer of funds to an external destination net of a configurable fee. Phase 2 handles reception of BEP-20 presale tokens, member validation, and pro-rata redistribution to contributors in proportion to their contributions.
- **Codebase:** Delivered as PerShare_SmartContracts_For_Audit.zip, containing the contract sources, Hardhat configuration, and package manifest.
Audit Type: Full manual security review supported by proof-of-concept exploit development on a local EVM.



AUDIT METHODOLOGY(1)

Project Familiarization

- Reviewed the contract's inline specification and the documented Phase 1 / Phase 2 model, including features explicitly deferred to V2 (cancel, pause-driven distribution, internal stablecoin distribution, vesting).
- Mapped the intended group-buy escrow flow (collect → collectively validate → send → receive presale tokens → redistribute) to the implemented code, noting where on-chain accounting and actual token balances diverge.

Automated Analysis

- Static analysis (Slither, Mythril) to surface reentrancy, integer overflow/underflow, uninitialized storage, gas inefficiencies, and unchecked external-call patterns.
- Compilation under the project's own settings (solc 0.8.20, viaIR enabled, optimizer at 200 runs) to confirm the audited bytecode matches the deployment configuration.

Manual Code Review

- Line-by-line review of [PerShareV1.sol](#).
- Verified member management and de-duplication, contribution accounting, validation/threshold logic, fund dispatch and fee calculation, expected-token handling, pro-rata token redistribution, and refund handling.
- Checked checks-effects-interactions ordering, [nonReentrant](#) coverage, rounding/dust in distribution math, fee-discount logic, and — critically — whether each payout path withdraws only funds that belong to the share being processed.

Scenario Simulation

- Compiled the contracts and exercised them against an in-memory EVM with proof-of-concept scripts driving real fund flows:
- Standard collection, collective validation, and dispatch to destination net of fee.
- Pro-rata Phase 2 distribution and post-deadline refunds.
- Multiple shares sharing the same stablecoin/token to test cross-share isolation.
- Malicious permissionless share creation pointing [expectedToken](#) at funds held for other shares.
- Blacklisted or reverting recipients inside the refund and distribution loops.
- Fee-on-transfer / non-standard token behaviour, and the post-deadline validate-vs-refund race.



AUDIT METHODOLOGY(2)

Risk Assessment

- Classified findings into Critical, High, Medium, Low, and Informational based on exploitability and protocol impact.
- Special focus on fund-custody and solvency risks — cross-share balance draining, payout-loop denial of service, and accounting mismatches that allow one share to be paid from another's deposits.

Tools Used

- Static analyzers: Slither, Mythril.
- Compilation: solc 0.8.20 (viaIR, optimizer 200 runs) via Hardhat.
- Scenario simulation: in-memory EVM (ganache) with ethers.js-driven PoC exploit scripts run against the compiled bytecode.
- Manual reasoning: Solidity 0.8.x checked-arithmetic guarantees and OpenZeppelin ReentrancyGuard, Pausable, and Ownable patterns.

Severity Classification

Severity	Definition
 Critical	Direct loss of user funds or protocol-level fund control is achievable by an external actor under realistic deployment conditions, with low cost and high reliability.
 High	Significant loss of funds is possible but requires specific preconditions, or causes systemic protocol malfunction without immediate fund loss.
 Medium	Limited loss of value, exploitable griefing, or economic logic deviations that meaningfully affect protocol behavior without direct fund theft.
 Low	Minor deviations from best practice, dead code, gas inefficiencies, or edge cases with negligible practical impact.
 Informational	Code quality observations, documentation gaps, or stylistic notes with no security or economic implication.



FINDINGS

■ Critical Risk

Cross-share distribution drains the entire contract balance

`validateDistribution` reads `IERC20(tokenAddress).balanceOf(address(this))` — the contract's **entire** balance of a token — and `_distributeTokens` pays **all of it** to one share's members. There is no per-share token accounting, and `setExpectedToken` accepts **any** address while `createShare` is **permissionless**. An attacker can create a throwaway share, point its `expectedToken` at the stablecoin (or any token the contract is custodying for others), and sweep it.

Impact:

Any user can **drain every other share's funds** under normal operating conditions. PoC result: an honest share holding **1,000 USDT** is emptied by an attacker who started with 100 USDT and ended with **1,099**; the victims' `refund()` then reverts. This breaks solvency completely.

Code (excerpt):

```
// setExpectedToken — accepts ANY token, incl. the stablecoin
s.expectedToken = tokenAddress; // L455

// validateDistribution — reads the GLOBAL balance, not this share's
uint256 tokenBalance =
    IERC20(tokenAddress).balanceOf(address(this)); // L345
require(tokenBalance > 0, "PerShare: no tokens received");
...
_distributeTokens(id, tokenAddress, tokenBalance); // L354

// _distributeTokens — splits that whole balance among one share
uint256 memberTokens = (totalTokens * contribution) / totalContrib; // L383
```

Recommendation:

Track tokens received **per share** and never distribute `balanceOf(address(this))`. Concretely:

- On the first Phase-2 deposit/validation, snapshot the amount and store `tokenReceived[id]`; distribute only that value.
- Reject `expectedToken == stablecoin` and reject a token already claimed by another active share.
- Emit the per-share distributed amount in `TokensDistributed`.



FINDINGS

■ High Risk

Reverting recipient locks refunds and distributions for the whole pool

`refund()` and `_distributeTokens()` both loop over every member and `require()` each individual transfer to succeed. If a **single** transfer reverts, the entire transaction reverts and **no one** is paid. USDT and USDC on BSC can **blacklist** addresses, and members may be contracts that revert on receipt — so this is externally triggerable, and also a deliberate griefing vector (a member makes themselves un-payable to freeze everyone else's money).

Impact:

One bad recipient **permanently locks the entire share's funds**. PoC result: two members each deposit 400 (800 in contract); after the deadline one member is blacklisted by the token issuer; the clean member's `refund()` **reverts** and **all 800 stays locked**. Affects both Phase-1 refunds and Phase-2 token distribution.

Code (excerpt):

```
// refund() – one failing transfer reverts the whole loop
for (uint256 i = 0; i < s.members.length; i++) {
    uint256 amount = contributions[id][member];
    if (amount > 0) {
        contributions[id][member] = 0;
        require(
            IERC20(s.stablecoin).transfer(member, amount), // L426
            "PerShare: refund failed"
        );
    }
}
// _distributeTokens() has the identical push-payment pattern // L386
```

Recommendation:

Replace push payments with a **pull pattern** so one member can never block another:

- Record each member's owed amount (`owedRefund[id][member]`, `owedTokens[id][member]`) and let each call `claimRefund(id)` / `claimDistribution(id)` to withdraw their own share.
- Alternatively, wrap each transfer so a single failure is skipped and recorded rather than reverting the batch.
- Apply `nonReentrant` to the claim functions and keep checks-effects-interactions ordering.



FINDINGS

■ Medium Risk

Fee-on-transfer / non-standard tokens over-credit contributions

`contribute()` credits `contributions[id][msg.sender]` and `s.totalReceived` using the **requested amount**, not the amount the contract actually received. For a fee-on-transfer or rebasing token, the contract records more than it holds. Because all shares share one commingled balance, the inflated share's later `_sendFunds/refund` can pull the shortfall from **other shares using the same token**. `stablecoin` is chosen per share and `createShare` is permissionless, so the token is not guaranteed to be a clean USDT/USDC.

Impact:

A share funded with a fee-on-transfer token records phantom deposits and pays them out from other shares' real balances, **draining honest pools** denominated in the same token and breaking solvency. Standard BSC USDT/USDC are not fee-on-transfer today, which is why this is Medium rather than High.

Code (excerpt):

```
require(
    IERC20(s.stablecoin).transferFrom(msg.sender, address(this), amount), // L260
    "PerShare: transfer failed"
);
contributions[id][msg.sender] += amount; // credits requested, not received
s.totalReceived                += amount; // L265-266
```

Recommendation:

Credit the **actual** amount received, measured by balance delta, and use `SafeERC20`:

```
uint256 before = IERC20(s.stablecoin).balanceOf(address(this));
IERC20(s.stablecoin).safeTransferFrom(msg.sender, address(this), amount);
uint256 received = IERC20(s.stablecoin).balanceOf(address(this)) - before;
contributions[id][msg.sender] += received;
s.totalReceived                += received;
```

Combined with per-share balance accounting (see Critical-1), this prevents any cross-share contamination. Optionally restrict `stablecoin` to a vetted allowlist.



FINDINGS

■ Medium Risk

validate() lacks a deadline check: post-deadline send/refund race

`validate()` never checks the deadline, while `refund()` requires the deadline to have passed. Once the deadline elapses **with the target met**, both paths are simultaneously open: a member can `validate()` to push the funds to the destination, or a member can `refund()` to pull the whole pool back to contributors. The outcome depends purely on who transacts first.

Impact:

The intended settlement becomes a **front-running race**. A single member can override the group's outcome — forcing a refund when quorum was about to send (stranding a legitimate presale purchase), or completing the send while others expect a refund. No direct theft, but the agreed fund flow is no longer deterministic, and the destination/contributors can be deprived of the result they reasonably expected.

Code (excerpt):

```
// validate() - no deadline guard
function validate(uint256 id) external onlyMember(id) notClosed(id) ... {
    require(s.totalReceived >= s.targetAmount, "PerShare: target not reached"); // L280
    ...                               // proceeds even after deadline
}

// refund() - only requires the deadline to have passed
require(block.timestamp > s.deadline, "PerShare: deadline not passed yet"); // L413
```

Recommendation:

Make the two paths mutually exclusive in time:

- Require `block.timestamp <= s.deadline` inside `validate()` so sends can only occur during the active window.
- Keep `refund()` gated on `block.timestamp > s.deadline` and on the share not already being sent (`notClosed` already covers `sent/refunded`).

This guarantees a clean cutover: validation/send before the deadline, refund only after.



FINDINGS

Low Risk

No recovery path for stuck or wrong Phase-2 tokens

Once `_distributeTokens` runs, `tokensDistributed` is set to `true` and distribution can never run again. If the presale never delivers, delivers the **wrong** token, or sends a **second tranche** after distribution, those tokens are stranded — there is no rescue or re-distribution function. Tokens accidentally sent to the contract address are likewise unrecoverable.

Impact:

Member funds can be **permanently locked** with no path to recover them. This is purely a lock (no theft), and it requires an external mistake or late delivery rather than an attacker, which is why it is rated Low — but for members the loss is total.

Code (excerpt):

```
function _distributeTokens(uint256 id, address tokenAddress, uint256 totalTokens) internal {
    Share storage s = shares[id];
    s.tokensDistributed = true;           // L369 – one-shot, no re-run, no rescue
    ...
}
// no function exists to recover a wrong token or a later tranche
```

Recommendation:

Add a guarded recovery function that, for a completed share, lets members claim a **later tranche** of the correct `expectedToken` (re-using the per-share accounting from Critical-1) rather than locking it.

Provide a narrow sweep for tokens that are **not** any share's `expectedToken` and not the stablecoin, so misdirected transfers can be returned.



FINDINGS

■ Informational

Distribution dust assigned to the first contributor

In `_distributeTokens`, pro-rata division leaves a small remainder ("dust") when the token amount doesn't divide evenly across members. This leftover is always sent to the first contributor in member-array order rather than being split fairly. The amount is negligible (a few token units) and poses no security or solvency risk — it's purely a minor fairness/predictability quirk.

Non-contributing members can reach quorum

Both `validate()` and `validateDistribution()` only check `isMember` — a member who contributed nothing can still cast a validation that counts toward the threshold. With a low threshold relative to the member count, quorum (and therefore the send or distribution) can be triggered entirely by members who never deposited. This is consistent with the M-of-N design, but worth flagging since voting weight is decoupled from funds at stake.

Duplicate member-getter functions

`getShareMembers` (L505) and `getMembers` (L538) are functionally identical — both simply return `shares[id].members`. The redundant second getter adds deployment cost and bytecode size with no benefit; one should be removed to keep the external interface clean.

Redundant zero-initialization of struct fields

In `createShare`, several `Share` fields are explicitly assigned their default values — `s.totalReceived = 0`, `s.sent = false`, `s.refunded = false`, `s.tokensDistributed = false`, `s.expectedToken = address(0)` (L234–238). Freshly allocated storage is already zero, so these writes only waste gas at creation time. They can be dropped without changing behavior.



CONCLUSION

The contract is **not safe for mainnet deployment in its current state**. The Critical finding allows any user, under normal operating conditions, to drain the entire balance the contract holds for every other share: confirmed by a working proof of concept in which an attacker converted 100 USDT into 1,099 by sweeping an honest 1,000 USDT pool. The High finding compounds the custody risk: a single blacklisted or reverting recipient permanently locks refunds and distributions for an entire share. Both stem from the same root cause — funds are commingled in one balance with no per-share token accounting, and payouts are pushed in loops rather than pulled.

The two Medium findings (fee-on-transfer over-crediting and the post-deadline validate/refund race) and the Low finding (no recovery path for stuck or wrong Phase-2 tokens) further affect solvency and predictability and should be resolved before launch. The informational items are code-quality and gas observations with no security impact.

On the positive side, the core contribution, validation, and send logic follows correct checks-effects-interactions ordering, **nonReentrant** is applied consistently across state-changing entry points, and the protocol fee is hard-capped; so the contract is free of the reentrancy and arithmetic classes of bugs. The issues identified are concentrated in fund custody and payout, not in control flow.

Recommendation: Implement per-share token accounting and convert distribution and refund to a pull-based pattern to resolve the Critical and High findings, then address the Medium and Low items. SpyWolf recommends a re-audit of the corrected code before any mainnet deployment. The development team was responsive and the codebase well-structured, and we expect a clean follow-up review once the custody logic is reworked.

This report reflects the state of the reviewed code at the time of audit and does not constitute financial advice or a guarantee against future vulnerabilities, particularly in code modified after this review.



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Disclaimer

This report shows findings based on our limited project analysis, following good industry practice from the date of this report, in relation to cybersecurity vulnerabilities and issues in the framework and algorithms based on smart contracts, overall social media and website presence and team transparency details of which are set out in this report. In order to get a full view of our analysis, it is crucial for you to read the full report.

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No applications were reviewed for security. No product code has been reviewed.